A. Course Handout (Version 1.0)

Institute/School Name	Chitkara University Institute of Engineering and Technology		
Department Name	Department of Applied Sciences		
Programme Name	Bachelor of Engineering (B.E.), Computer Science & Engineering		
Course Name	Discrete Structures	Session	2024-2025
Course Code	22AS003	Semester/Batch	4th /2023
L-T-P (Per Week)	4-0-0	Course Credits	04
Pre-requisite	+2 Level Mathematics, Combinatorics	NHEQF Level	5
Course Coordinator	Dr. Madhu Aneja	SDG Number	9

CLO01	To apply the knowledge obtained to investigate and solve a variety of live problems.
CLO02	Analysis of relevance within the context of computer science, in the areas of data structures and algorithms
CLO03	To understand and apply the theory and techniques of Lattice, Logic and Boolean algebra.
CLO04	To comprehend Graph Theory and its relevance within the context of computer science and finding solutions of live problems related to shortest path etc.

1. Objectives of the Course

The course provides the ability to develop the mathematical foundations, abstraction and formalization of reasoning for the problems related to computer science engineering to arrive at substantiated conclusions.
The main objectives of the courses are:

- To understand the theory and techniques of logic, graphs and trees, and algebraic systems.
- To apply the knowledge and skills obtained to investigate and solve a variety of discrete mathematical problems.
- To communicate mathematical ideas and to make effective use of appropriate technology.
- To comprehend discrete structures and their relevance within the context of computer science, in the areas of data structures and algorithms, in particular (Enabling Knowledge).
- To apply discrete structures into other computing problems such as formal specification, verification, databases, and cryptography (Problem Solving).

2. Course Learning Outcomes

After completion of the course, student should be able to:

	Course Outcome	POs, PSOs	**CL	***KC	Lectures
CLO01	To apply the knowledge obtained to investigate and solve a variety of live problems.	PO1,PO2, PSO1	K2	Factual Conceptual	**15**
CLO02	Analysis of relevance within the context of computer science, in the areas of data structures and algorithms	PO1,PO2, PSO1	K3	Conceptual Procedural	**15**

CLO03	To understand and apply the theory and techniques of Lattice, Logic and Boolean algebra.	PO2,PO3, PSO2, PSO3	K3	Conceptual Procedural	**11**
CLO04	To comprehend Graph Theory and its relevance within the context of computer science and finding solutions of live problems related to shortest path etc.	PO1,PO2, P03, PSO2, PSO3	K3	Conceptual Procedural	**7**
Total Contact Hours					**48**

Revised Bloom's Taxonomy Terminology

* PO's available at (shorturl.at/cryzF)

**Cognitive Level =CL

***Knowledge Categories = KC

Course Learning Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PS 01**	**PS 02**	**PS 03**
CLO01	H	H									H		M		
CLO02	H	H											M		
CLO03		H	M											M	H
CLO04	H	M	H											M	H

H=High, M=Medium, L=Low

3. ERISE Grid Mapping

Feature Enablement	**Level (1-5, 5 being highest)**
Entrepreneurship	2
Research	3
Innovation	2
Skills	3
Employability	4

4. Recommended Books:

B01: Elements of Discrete Mathematics, by C.L.Liu, McGraw-Hill.

B02: Discrete Mathematics by Babu Ram, Pearson Education India.

B03: Discrete Mathematics by Lipschutz/Lipson, Schaum Series, 2nd Edition, TMH, Edition.

B04: Discrete Mathematics and its Applications by Kenneth H. Rosen, McGraw- Hill.

B05: Logic and Discrete Mathematics by Trembly/Grassmann, Pearson Education.

B06: The Discrete Mathematics, Chitkara University Publications.

E-Resources:

https://library.chitkara.edu.in/subscribed-books.php

5. Other readings & relevant websites:

Serial No	**Link of Journals, Magazines, websites and Research Papers**
1.	http://download.nos.org/srsec311new/L.No.15-A.pdf

2.	http://faculty.simpson.edu/lydia.sinapova/www/cmsc180/LN180_Johnsonbaugh-07/L17-EquivalenceRel.htm
3.	http://web.cs.wpi.edu/~cs504/s00m/notes/recurrence/solve/step2/step2.html
4.	http://www.iep.utm.edu/prop-log/
5.	http://plato.stanford.edu/entries/boolalg-math/
6.	http://www.hamilton.ie/ollie/Downloads/Graph.pdf

6. Recommended Tools and Platforms

MatLab, Wolfram Mathematica, Python and Sage.

7. Course Plan:

Lecture Number	Topics	Recommended Book
1-2	**Recurrence relation**: Homogeneous and Non-homogeneous recurrence relation with constant coefficients.	B01, B02
3-5	**Characteristic polynomial & Introduction to generating functions:** Methods of generating functions.	B01, B02
6-8	**Logic**: Propositions & logical operators, Truth table. Laws of logic & Quantifiers, Validity of arguments, Truth values.	B01, B06
9-12	**Lattices**: Definition of lattice, Sub lattice, Isomorphic lattices, Bounded lattice, Distributive lattice, Complemented lattice, Modular lattice.	B01, B02
13-15	**Boolean Algebra**: Atoms, Boolean expressions, Applications of Boolean algebra to switching theory & Logic Gates.	B01, B06
Sessional Test-1 (1– 15 Lectures)		
16-19	**Graph Theory:** Introduction to graphs & Sub graphs. Directed & Undirected graph, Order& Size of graph, Degree of vertex, Source, sink, Eccentricity, Trail, Walk, Path, Distance, Diameter, Cycle, Wheel.	B02
20-24	Multi graph, Planer graph, Pseudo graph, weighted graph, Regular, Complete and Traversable graph. Isomorphism of graphs,	B06
25-26	Sub graphs, Complement of graph, Adjacency Matrix, Adjacency List, Incidence Matrix.	B06
27-30	Complete Bipartite graph & Spanning Graph, Cut vertex, Cut edge, Region, Eulerian & Hamiltonian graphs.	B03
31-32	**Euler's formula & its applications:** Traveling salesman problem, Konigsberg Bridge problem.	B02
33-34	Chromatic number & Graph coloring by Welsh Powell Algorithm.	B02
Sessional Test-2 (1–34 Lectures)		
35-36	**Tree**: Binary trees, Traversing binary trees.	B06
37-38	Rooted & spanning tree, Algebraic expression trees.	B06
39-40	Depth-First Search Algorithm, Breadth-First Search Algorithm.	B06
41-45	**Shortest path algorithm:** Kruskal's Algorithm, Dijkstra's Algorithm.	B05
46-48	Warshall's Algorithm, Prim's Algorithm.	B05

ETE Syllabus (Lecture number 1-48)

8. Delivery/Instructional Resources

Lecture No.	Topics	Web References	Audio-Video
1-2	**Recurrence relation**: Homogeneous and Non-homogeneous recurrence relation with constant coefficients.	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
3-5	**Characteristic polynomial & Introduction to generating functions:** Methods of generating functions.	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
6-8	**Logic**: Propositions & logical operators, Truth table. Laws of logic & Quantifiers, Validity of arguments, Truth values.	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
9-12	**Lattices**: Definition of lattice, Sub lattice, Isomorphic lattices, Bounded lattice, Distributive lattice, Complemented lattice, Modular lattice.	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
13-15	**Boolean Algebra**: Atoms, Boolean expressions, Applications of Boolean algebra to switching theory & Logic Gates.	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
16-19	**Graph Theory:** Introduction to graphs & Sub graphs. Directed & Undirected graph, Order& Size of graph, Degree of vertex, Source, sink, Eccentricity, Trail, Walk, Path, Distance, Diameter, Cycle, Wheel.	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
20-24	Multi graph, Planer graph, Pseudo graph, weighted graph, Regular, Complete and Traversable graph. Isomorphism of graphs,	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
25-26	Sub graphs, Complement of graph, Adjacency Matrix, Adjacency List, Incidence Matrix	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/

27-30	Complete Bipartite graph & Spanning Graph, Cut vertex, Cut edge, Region, Eulerian & Hamiltonian graphs	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
31-32	**Euler's formula & its applications:** Traveling salesman problem, Konigsberg Bridge problem, `	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
33-34	Chromatic number & Graph coloring by Welsh Powell Algorithm.	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
35-36	**Tree**: Binary trees, Traversing binary trees,	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
37-38	Rooted & spanning tree, Algebraic expression trees,	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
39-40	Depth-First Search Algorithm, Breadth-First Search Algorithm.	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
41-45	**Shortest path algorithm:** Kruskal's Algorithm, Dijkstra's Algorithm	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
46-48	Warshall's Algorithm, Prim's Algorithm	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/

9. Action plan for different types of learners

Slow Learners	Average Learners	Fast Learners
Multiple remedial Extra classes Encouragement for improvement using Peer Tutoring.	Doubt-sessions, and practice questions.	Quiz Competitions, MCQ, and practice questions

10. Evaluation Scheme & Components:

Evaluation Component	Type of Component	No. of Assessments	Weightage of Component	Mode of Assessments
Component 1	Subjective Test/ Sessional Tests (STs)	02*	50%	Online/Offline
Component 2	End Term Examinations	01**	50%	Online
Total		**100%**		

* Best out of ST1 and ST2

** As per Academic Guidelines minimum 75% attendance is required to become eligible for appearing in the End Semester Examination

11. Syllabus of the Course:

Subject: Discrete Structures	**Subject Code: 22AS003**

S. No.	Topic (s)	No. of Lectures	Weightage %
1	**Recurrence relation**: Homogeneous and Non-homogeneous recurrence relation with constant coefficients, **Characteristic polynomial & Introduction to generating functions:** Methods of generating functions. **Logic**: Propositions & logical operators, Truth table. Laws of logic & Quantifiers, Validity of arguments, Truth values. **Lattices**: Definition of lattice, Sub lattice, Isomorphic lattices, Bounded lattice, Distributive lattice, complemented lattice, Modular lattice. **Boolean Algebra**: Atoms, Boolean Sex pressions, Applications of Boolean algebra to switching theory & Logic Gates.	**15**	**30%**

2	**Graph Theory:** Introduction to graphs & Sub graphs. Directed & Undirected graph, Order& Size of graph, Degree of vertex, Source, sink, Eccentricity, Trail, Walk, Path, Distance, Diameter, Cycle, Wheel., Multi graph, Planer graph, Pseudo graph, weighted graph, Regular, Complete and Traversable graph. Isomorphism of graphs, Sub graphs, Complement of graph, Adjacency Matrix, Adjacency List, Incidence Matrix, Complete Bipartite graph & Spanning Graph, Cut vertex, Cut edge, Region, Eulerian & Hamiltonian graphs. **Euler's formula & its applications:** Traveling salesman problem, Konigsberg Bridge problem, Chromatic number & Graph coloring by Welsh Powell Algorithm, Rooted & spanning tree, Algebraic expression trees, Depth-First Search Algorithm, Breadth-First Search Algorithm. **Tree**: Binary trees, Traversing binary trees, Boolean expressions, Applications of Boolean algebra to switching theory & Logic Gates Applications of Boolean algebra to switching theory & Logic Gates.	**26**	**55%**
3	**Shortest path Algorithm:** Kruskal's Algorithm, Dijkstra's Algorithm, Warshall's Algorithm, Prim's Algorithm	**7**	**15%**

This Document is approved by:

Designation	**Name**	**Signature**
Course Coordinator	Dr. Madhu Aneja	
Program Head	Dr. Reetu Malhotra	
Dean	Dr. Mohit Kumar Kakkar	
Date (DD/MM/YYYY)	30/11/2024	